

SLIIT
SE3082 – Parallel Computing



Lab 7
Exercise 03

Akash D
IT24101853

Exercise 03: Scatter + Gather

Implementation Code (sum_gather.c)

```
#include <mpi.h>
#include <stdio.h>
#include <stdlib.h>
#define N 1000000

int main(int argc, char **argv) {
    MPI_Init(&argc, &argv);

    int rank, size;

    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    int chunk_size = N / size;

    int *array = NULL;

    if (rank == 0) {
        array = (int *)malloc(N * sizeof(int));
        for (int i = 0; i < N; i++) array[i] = i + 1;
    }

    int *local_chunk = (int *)malloc(chunk_size * sizeof(int));

    double start = MPI_Wtime();

    MPI_Scatter(array, chunk_size, MPI_INT, local_chunk, chunk_size, MPI_INT, 0,
        MPI_COMM_WORLD);

    long long local_sum = 0;

    for (int i = 0; i < chunk_size; i++) local_sum += local_chunk[i];

    long long *all_sums = NULL;

    if (rank == 0) all_sums = (long long *)malloc(size * sizeof(long long));

    MPI_Gather(&local_sum, 1, MPI_LONG_LONG, all_sums, 1, MPI_LONG_LONG, 0,
        MPI_COMM_WORLD);

    if (rank == 0) {
        long long total_sum = 0;

        for (int i = 0; i < size; i++) total_sum += all_sums[i];
```

```
double elapsed = MPI_Wtime() - start;
printf("\n[Ex3 Gather] Total sum: %lld, Time: %.4f sec\n", total_sum, elapsed);
free(all_sums);
}
if (rank == 0) free(array);
free(local_chunk);
MPI_Finalize();
return 0;
}
```

Test Output (4 processes)

[Ex3 Gather] Total sum: 500000500000, Time: 0.0048 sec